

The Stationary Wave Equation

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Separation of the Wave Equation

Let us start by assuming that the potential energy of systems we are interested in are a function only of spatial position.

$$V(\vec{r}, t) = V(\vec{r})$$

Thus we can write

$$i\hbar \frac{\partial \Psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\vec{r}, t) + V(\vec{r}) \Psi(\vec{r}, t)$$

We seek solutions subject to initial and boundary conditions

Separation of the Wave Equation

Since V is a function of space only, we can use separation of variables

$$\Psi(\vec{r}, t) = \phi(t)\psi(\vec{r})$$

Substituting this into the wave equation and dividing by Ψ

$$\frac{i\hbar}{\phi} \frac{\partial \phi}{\partial t} = \frac{1}{\psi} \left[-\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r})\psi \right]$$

Separation of the Wave Equation

Since the two sides are functions of different variables, they must be equal to within a constant

$$\frac{i\hbar}{\phi} \frac{\partial \phi}{\partial t} = C$$

$$\frac{1}{\psi} \left[-\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r})\psi \right] = C$$

Separation of the Wave Equation

Solving the time dependent part

$$i \frac{\partial \phi}{\partial t} = \frac{C}{\hbar} \phi \quad \longrightarrow \quad \phi(t) = \exp\left(-i \frac{C}{\hbar} t\right)$$

Therefore

$$\Psi(\vec{r}, t) = \psi(\vec{r}) \exp\left(-i \frac{C}{\hbar} t\right)$$

The Energy

Let us use this to calculate the expectation value of the energy

$$\langle \varepsilon \rangle = \int_V \Psi^* \left(i\hbar \frac{\partial}{\partial t} \right) \Psi dV = \int_V \psi^* e^{i\frac{C}{\hbar}t} C \psi e^{-i\frac{C}{\hbar}t} dV = C$$

Thus, the separation constant is equal to the energy of the system

The Stationary Wave Equation

It remains to solve the Stationary Wave Equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\vec{r})\psi = \varepsilon\psi$$

Subject to appropriate boundary conditions and knowledge of $V(r)$

As we shall see, this is an eigen-equation and leads to solutions only for discrete values of the energy

Two Particle Systems

For two particles the total energy is

$$\varepsilon = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(\vec{r})$$

If we transformed the wave equation into center of mass coordinates

$$\frac{\hbar^2}{2m_t} \nabla^2 \psi_e + \varepsilon_e \psi_e = 0 \quad \text{External motion}$$

$$\frac{\hbar^2}{2\mu} \nabla^2 \psi_{\text{int}} + (\varepsilon_{\text{int}} - V(\vec{r})) \psi_{\text{int}} = 0 \quad \text{Internal motion}$$

Two Particle Systems

where

$$\mathcal{E} = \mathcal{E}_e + \mathcal{E}_{\text{int}}$$

$$m_t = m_1 + m_2$$

and

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \quad \text{Is called the reduced mass}$$

Note: we have assumed no potential acts on the particle as a whole